

NIRAJ CHAUHAN

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PROFESSIONAL SUMMARY

Data Engineer with 1.5+ years of experience building scalable data extraction and ETL pipelines, processing 15M+ records across global platforms. Skilled in web data acquisition, data cleaning, validation, and database optimization using Python and MySQL. Hands-on experience across domains including e-commerce, quick-commerce, industrial automation, real estate, travel & hospitality, and global trade & certification datasets. Proven ability to design reliable, automated data workflows with scheduling, retry mechanisms, and proxy management to ensure high-quality structured data for analytics and business intelligence

TECHNICAL SKILLS

Programming: Python

Web Data Extraction: Scrapy, Selenium, Playwright, BeautifulSoup, Requests

Data Engineering: ETL Pipelines, Data Cleaning, Data Validation, Data Transformation, Batch Processing

Data Processing: Pandas, NumPy

Databases: MySQL, MongoDB (Indexing, Query Optimization)

Automation: Windows Task Scheduler, Cron Jobs (basic), Scheduling Pipelines

Core Concepts: Data Cleaning, Deduplication, Data Normalization, Batching, Proxy Rotation, Retry Logic

System Reliability: Proxy Rotation, Retry Logic, Error Handling, Logging

Tools: Git, GitrHub,Linux Basics

WORK EXPERIENCE

Data Engineer

Purra.ai | July 2025 – April 2026

- Engineered large-scale data acquisition and ETL pipelines processing 15M+ trade and certification records across multiple international platforms
- Designed scalable crawler architecture for dynamic, search-based, and paginated registry platforms, ensuring high reliability
- Extracted structured datasets from company directories (US, Canada, India), government contractor databases (CAGE codes), and global certification registries (Halal, Kosher, FSC, BRCGS, OEKO-TEX)
- Implemented batch processing, retry logic, proxy rotation, and structured logging, improving fault tolerance and reducing failure rates
- Performed large-scale data cleaning, normalization, and deduplication on 1.5+ crore records to ensure data accuracy and consistency
- Optimized database insert operations and indexing strategies, improving large-scale data storage and query performance
- Built automated data workflows ensuring reliable and consistent data ingestion for analytics platforms

Python Developer – Web Scraping & Data Pipelines

Actowiz Solutions| September 2024 – July 2025

- Developed automated web scraping pipelines for e-commerce, quick-commerce, pharmaceutical, restaurant, and retail platforms
- Extracted structured product data including pricing, ratings, stock availability, and seller information at scale
- Built data extraction workflows for industrial automation suppliers and B2B manufacturing catalogs
- Handled JavaScript-rendered platforms using Selenium and Playwright for dynamic data extraction

- Designed and automated scraping workflows using Python and Windows Task Scheduler, reducing manual effort and enabling scheduled data collection
- Stored and managed structured datasets in MySQL and MongoDB with optimized schema design
- Improved parsing logic and selector strategies, increasing scraping efficiency and reducing execution time
- Contributed to real estate and cruise industry data scraping projects

EDUCATION

- Bachelor of Computer Applications (BCA) 2024
Lj College of Computer Application

KEY PROJECT HIGHLIGHTS

End-to-End E-commerce Data Pipeline

- Designed and built an automated ETL pipeline to extract product data (price, ratings, availability) from e-commerce platforms
- Implemented data extraction using Scrapy and Requests with retry logic and proxy rotation to handle anti-bot mechanisms
- Performed data cleaning, validation, and transformation using Pandas to ensure high data quality
- Designed optimized MySQL schema with indexing for efficient storage and retrieval
- Automated pipeline execution using Windows Task Scheduler, enabling scheduled data ingestion
- Processed and stored large-scale datasets for analytics and reporting

Global Trade & Certification Data System (15M+ Records)

- Engineered large-scale data acquisition system to extract and process 15M+ global trade and certification records
- Extracted structured data from company directories, government databases, and certification platforms (Halal, Kosher, FSC, BRCGS, OEKO-TEX)
- Implemented data normalization, deduplication, and validation pipelines to ensure consistency across multiple data sources
- Built scalable data workflows with batch processing, logging, and retry mechanisms for fault tolerance
- Optimized database insertion and indexing strategies, improving performance for large datasets
- Delivered clean, structured datasets for business intelligence and analytics use cases

Dynamic Website Data Extraction System

- Built robust scraping system for JavaScript-heavy and dynamic websites using Selenium and Playwright
- Handled anti-bot challenges using headers, cookies, session management, and proxy rotation
- Implemented scrolling, clicking, and human-like interaction patterns to reduce detection
- Extracted structured data from multiple domains including real estate, travel, and B2B platforms
- Improved scraping success rate and reduced blocking issues through optimized request handling